



FP1 Chapter 8 Matrices (2)

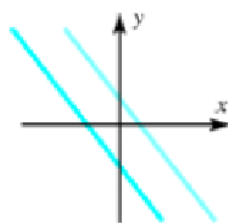
**Lecture 8 Matrices (2)**

(System of Linear Equations, Eigenvalue, Eigenvector, Diagonalisation, Characteristic Equations)

**(A) System and Consistency of Linear Equations**

**(I) Two variables**

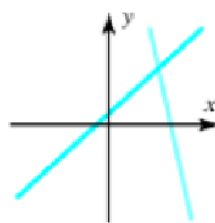
- A system of equation is called **consistent** if there is at least one solution. In this case, the system may have one solution or infinite solution.
- If there are no solutions to the system it is known to be **inconsistent**.
- Generally, only system of  $n$  linear equation in  $n$  variables will have a **unique** solution.
- Also when a system of linear equations is written in the form  $Ax = b$ ,  $\det(A)$  cannot equal to 0 for the system to have a unique solution.
- A system of linear equations in two variables (2 equations in the form  $a_1x_1 + a_2x_2 = b$ ) are a pair of straight lines. Only 3 possible cases exist:



No solution

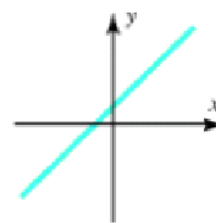
$$\begin{aligned} x + y &= 3 - (1) \\ 3x + 3y &= 6 - (2) \\ (1) \times 3: 9 &= 6 \end{aligned}$$

No solutions.



One solution

$$\begin{aligned} x - y &= 1 - (1) \\ x + y &= 5 - (2) \\ (1) + (2): 2x &= 6 \\ x = 3, y &= 5 - 3 = 2 \end{aligned}$$



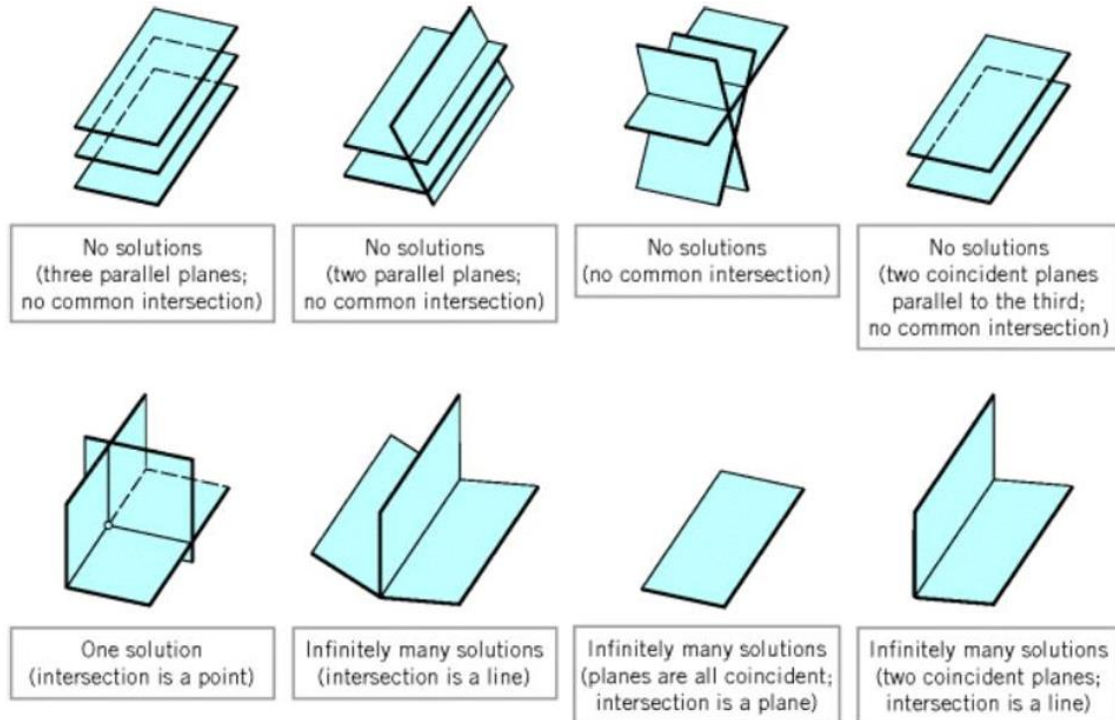
Infinitely many solutions  
(coincident lines)

$$\begin{aligned} x - y &= 1 - (1) \\ 4x - 4y &= 4 - (2) \\ (1) \times 4 &= (2) \end{aligned}$$

Let  $x = t, y = t - 1$ .  
The solution is a line.

## (II) Three variables

- A system of linear equations in three variables represent the equation of multiple planes. Many cases can occur as shown below:



- In cases of parallel planes, the coefficients, which form the normal vectors to the planes, will be multiple of each other.

## System of Linear Equations

- Many problems in science, engineering, economics and other fields can be described mathematically by a system of equations. The solutions of the system of equations provide insight into the solutions of the practical problem.
- We shall study a general method (or algorithm) for solving a system of linear equations in  $n$  unknowns.
- The general form of a system of linear equations is  $Ax = b$ . If  $A^{-1}$  exists, we can multiply with  $A^{-1}$ .

$$Ax = b$$

$$A^{-1}Ax = A^{-1}b \Rightarrow Ix = A^{-1}b \Rightarrow x = A^{-1}b$$

- **Example 1:**

Solve the system of equations below:

$$x + 4y - z = 1 \quad -2x + y + 2z = 2 \quad 3x + 5y - z = 5$$

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- The inverse matrix method cannot solve systems that have a matrix with determinant zero. To solve these types of equations, we can write the system into an augmented matrix and reduce it using elementary row operations. For example,

- 4

- The following example shows the elementary row operation to solve a system of equations.

- For a system of equations, where the reduced augmented matrix is of the form

$$\left( \begin{array}{cccc|c} * & * & * & : & * \\ * & * & * & : & * \\ 0 & 0 & \alpha & : & \beta \end{array} \right),$$

there are three cases to consider:

- (i) If  $\alpha = 0, \beta = 0$ , then there is an **infinite number of solutions**. ( $\det = 0$ )
- (ii) If  $\alpha = 0$  and  $\beta \neq 0$ , then there are **no solutions**. ( $\det = 0$ )
- (iii) If  $\alpha \neq 0$  and  $\beta \in \mathbb{R}$ , there will be a **unique solution**. ( $\det \neq 0$ )

- **Example 2:**

Find the set of values of  $a$  for which the system of equations

$$ax + y + 2z = 0 \quad 3x - 2y = 4 \quad 3x - 4y - 6az = 14$$

has a unique solution.

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- **Example 3:**

Determine the values of  $k$  for which the system of equations

$$x + 2y + 3z = 4 \quad 2x + 3y + kz = 9 \quad x + ky + 6z = 1$$

has no unique solution.

Show that the system is consistent for one of these values of  $k$  and inconsistent for the other.

- **Example 4:**

It is given that the matrix

$$M = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 4 & a \\ 9 & -5 & -7 \end{pmatrix}$$

is non-singular.

- (i) Find the set of possible values of  $a$
- (ii) Find the inverse of  $M$ .
- (iii) Solve the equations

$$2x - z = 1 \quad 4y + az = -3 \quad 9x - 5y - 7z = 1$$

for  $x, y, z$  in terms of  $a$ .



- **Example 5:**

Find the set of values of  $a$  for which the system of equations

$$x + 4y + 12z = 5 \quad 2x + ay + 12z = a - 1 \quad 3x + 12y + 2az = 10$$

has a unique solution.

Show that the system does not have any solution in the case  $a = 18$ .

Given that  $a = 8$ , show that the number of solutions is infinite and find the solution for which  $x + y + z = 1$ .

- **Example 6:**

Given that

$$M = \begin{pmatrix} 1 & 5 & a \\ 2 & -4 & 1 \\ 4 & 6 & 7 \end{pmatrix},$$

evaluate  $\det(M)$  in terms of  $a$ .

Hence show that, whatever the values of the constants  $p, q$  and  $r$ , the equations

$$x + 5y + az = p \quad 2x - 4y - z = q \quad 4x + 6y + 7z = r$$

has a unique solution provided  $a \neq 3$ .

For the case where  $M$  is non-singular, find  $M^{-1}$  and hence find, in terms of  $a$ , the solution of the equation where  $p = 1, q = 0, r = -2$ .

Interpret the equations geometrically when  $a = 3, p = -18, q = 7$  and  $r = -29$ .



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- **Example 7:**

Find the value of  $k$  for which the system of equations

$$x + 3y + kz = 4 \quad 4x - 2y - 10z = -5 \quad x + y + 2z = 1$$

has no unique solution.

For this value of  $k$ , find the set of possible solutions, giving your answer in the form

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = a + tb,$$

where  $a$  and  $b$  are vectors and  $t$  is a scalar.



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- **Example 8:**

Find the value of  $a$  for which the system of equations

$$x - y + 2z = 4 \quad x + ay - 3z = b \quad x - y + 7z = 13$$

where  $a$  and  $b$  are constants, has no unique solution.

Taking  $a$  as the value just found,

- (a) find the general solution in the case  $b = -5$ .
- (b) interpret the solution geometrically in the case  $b \neq -5$ .

- **Example 9 (PYQ M/J 2012/13/10):**

Find the set of values of  $a$  for which the system of equations

$$\begin{aligned}x - 2y - 2z &= -7, \\2x + (a - 9)y - 10z &= -11, \\3x - 6y + 2az &= -29,\end{aligned}$$

has a unique solution. [4]

Show that the system has no solution in the case  $a = -3$ . [2]

Given that  $a = 5$ ,

(i) show that the number of solutions is infinite, [2]

(ii) find the solution for which  $x + y + z = 2$ . [3]

- **Exercise A:**

1. Solve the system of equations below:

$$x - 2y + z = 1 \quad x + y - 2z = 1 \quad 2x - y - z = -1$$

2. Solve the system of equations below:

$$x - 2y + z = 1 \quad 2x - 4y + 2z = 3 \quad 2x - y - z = -1$$

3. Solve the system of equations below:

$$x - 2y + z = 1 \quad 2x - 4y + 2z = 2 \quad 2x - y - z = -1$$

4. Solve the system of equations below:

$$x - 2y + z = 1 \quad 2x - 4y + 2z = 3 \quad 3x - 6y + 3z = 5$$

5. Solve the system of equations below:

$$ax + y + z = 1 \quad x + ay + z = a \quad x + y + az = a^2$$

6. Find the values of  $\lambda$  and the related values of  $k$  so that the system of equations

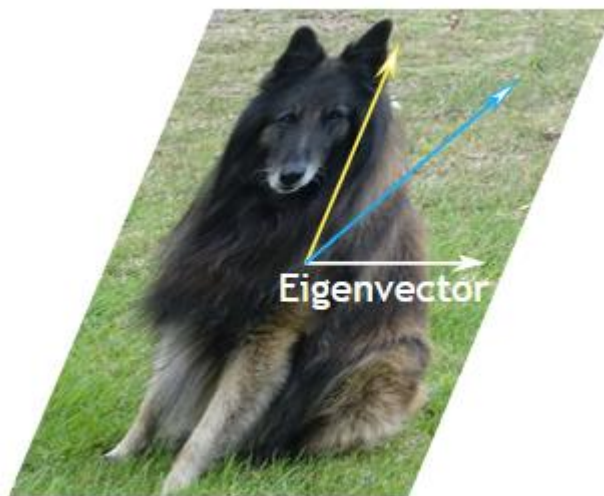
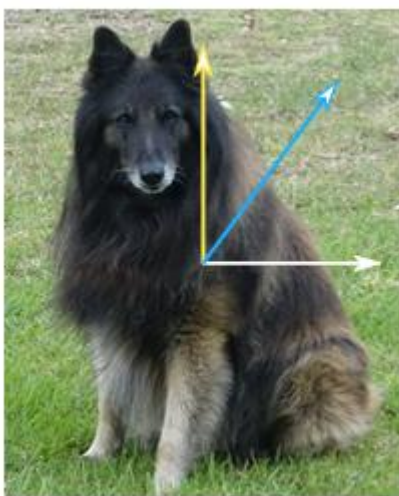
$$4x + y - 6z = 8 \quad \lambda x - y - z = 13 \quad x - 2y + \lambda z = k$$

- (i) has unique solution,
- (ii) has an infinite number of solutions,
- (iii) has no solutions.
- (iv) If  $\lambda = 4$  and  $k = 14$ , find the solutions of the equations.
- (v) If  $\lambda = 3$  and  $k = 11$ , find the solutions of the equations.



## (B) Eigenvalues and eigenvectors

- If  $A$  is an  $m \times n$  matrix and  $x$  is a vector in  $\mathbb{R}^n$ , then there is usually no geometric relationship between the vector  $x$  and the vector  $Ax$ . However, there are often certain non-zero vectors  $x$  and  $Ax$  are scalar multiples of one another.
- Such vectors arise naturally in the study of vibrations, electrical systems, genetics, chemical reactions, quantum mechanics, mechanical stress, economics and geometry.
- The equation  $Ax = b$ , which is investigated in previous sections. The equation  $Ax = \lambda x$ , which is what we will investigate here.
- The word 'Eigen' is of German origin and means 'own' - belonging to or ownership. In Mathematics we adapt its meaning to mean characteristic. So there are **characteristics values** and **characteristics equations** of the particular matrix we are studying.
- If  $A$  is an  $n \times n$  matrix, then a **non-zero** vector  $x$  in  $\mathbb{R}^n$  is called an eigenvector of  $A$  if  $Ax$  is a scalar multiple of  $x$ ; that is,  $Ax = \lambda x$  for some scalar  $\lambda$ . The scalar  $\lambda$  is called an **eigenvalue** of  $A$ , and  $x$  is said to be **eigenvector** of  $A$  **corresponding** to  $\lambda$ .
- Eigenvector **does not change direction** in a transformation.



- While eigenvalue is the scale of the stretch:



1 means no change; 2 means doubling in length,  $-1$  means pointing backwards along the eigenvalue's direction.

- For example, given matrix  $\begin{pmatrix} -6 & 3 \\ 4 & 5 \end{pmatrix}$ , an eigenvector is  $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$  with a matching eigenvalue of 6. Let's verify with formula  $Ax = \lambda x$ .

- In short, this equation literally describes vectors  $x$  which are only scaled when they are transformed using the matrix  $A$ .
- To find these eigen things, we will solve this system by introducing the identity matrix  $I$ , since when we factor out matrix  $X$ , the difference between a matrix and a scalar cannot be evaluated.

$$Ax - \lambda x = 0$$

$$Ax - \lambda Ix = 0$$

$$x(A - \lambda I) = 0$$

$$x = (A - \lambda I)^{-1} 0 = 0$$

- For  $\lambda$  to be an eigenvalue, there must be a non-zero solution of this equation. However, this equation has a non-zero solution if and if

$$\det(A - \lambda I) = 0.$$

- This is known as the characteristic equation / polynomial of  $A$ . The scalars satisfying this equation are the eigenvalues of  $A$ . When expanded, the determinant  $\det(A - \lambda I) = 0$  is a polynomial equation in  $\lambda$ .
- It can be shown that if  $A$  is  $n \times n$  matrix, then the characteristics equation of  $A$  has degree  $n$ . For Further Maths,  $n$  is either 2 or 3.

- **Proving:**

1. If  $\lambda$  is an eigenvalue of  $A$  and  $x$  is a corresponding eigenvector, show that  $\lambda^2$  is an eigenvalue of  $A^2$  and  $x$  is a corresponding eigenvector.

2. If  $k$  is a positive integer,  $\lambda$  is an eigenvalue of  $A$ , and  $x$  is a corresponding eigenvector, then  $\lambda^k$  is an eigenvalue of  $A^k$  and  $x$  is a corresponding eigenvector.

3. If  $\lambda$  is an eigenvalue of  $A$  and  $x$  is a corresponding eigenvector, prove that  $k\lambda$  is an eigenvalue of  $kA$ , and  $x$  is a corresponding eigenvector.

4. If  $\lambda$  is an eigenvalue of  $\mathbf{A}$  and  $\mathbf{x}$  is a corresponding eigenvector, prove that  $\lambda^{-1}$  is an eigenvalue of  $\mathbf{A}^{-1}$ , and  $\mathbf{x}$  is a corresponding eigenvector.
5. If  $\lambda$  is an eigenvalue of  $\mathbf{A}$  and  $\mathbf{x}$  is a corresponding eigenvector, and  $s$  is a scalar, prove that  $\lambda - s$  is an eigenvalue of  $\mathbf{A} - s\mathbf{I}$ , and  $\mathbf{x}$  is a corresponding eigenvector.
6. If  $\mathbf{x}$  is an eigenvector of each of the matrices  $\mathbf{A}$  and  $\mathbf{B}$ , with corresponding eigenvalues  $\lambda$  and  $\mu$  respectively, prove that  $\lambda\mu$  is an eigenvalue of  $\mathbf{AB}$ , and  $\mathbf{x}$  is a corresponding eigenvector.

7. If  $x$  is an eigenvector of each of the matrices  $A$  and  $B$ , with corresponding eigenvalues  $\lambda$  and  $\mu$  respectively, prove that  $\lambda + \mu$  is an eigenvalue of  $A + B$ , and  $x$  is a corresponding eigenvector.
8. The matrix  $A$  has  $\lambda$  as an eigenvalue with  $e$  as a corresponding eigenvector. Show that if  $A$  is non-singular then  $\lambda \neq 0$ .

**Case 1:  $2 \times 2$  matrix**

- The characteristic equation  $\det(A - \lambda I) = 0$  is a quadratic equation. Only real and distinct eigenvalues are considered.
- An eigenvector is a solution of  $Ax = \lambda x$ . Hence to find an eigenvector corresponding to  $\lambda = \lambda_1$ , you have to find a column vector  $\begin{pmatrix} x \\ y \end{pmatrix}$  satisfying the equation  $Ax = \lambda_1 x$ .

- **Example 10:**

Find the eigenvalues of the matrix

$$\begin{pmatrix} 4 & -2 \\ 3 & -1 \end{pmatrix}.$$

Find also the corresponding eigenvectors.

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**Case 2:  $3 \times 3$  matrix**

- The characteristic equation  $\det(A - \lambda I) = 0$  is a cubic equation. Only real and distinct eigenvalues are considered.
- An eigenvector is a solution  $Ax = \lambda x$ . Hence to find an eigenvector corresponding to  $\lambda = \lambda_1$ , you have to find a column vector  $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$  satisfying the equation  $Ax = \lambda x$ .

- **Example 11:**

It is given that

$$A = \begin{pmatrix} 1 & -1 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & -3 \end{pmatrix}.$$

Write down the eigenvalues of  $A$  and find corresponding eigenvectors.



- **Example 12 (PYQ O/N 2008/1/4):**

The matrix  $A$  has  $\lambda$  as an eigenvalue with  $e$  as a corresponding eigenvector. Show that  $e$  is an eigenvector of  $A^2$  and state the corresponding eigenvalue. [3]

Given that one eigenvalue of  $A$  is 3, find an eigenvalue of the matrix  $A^4 + 3A^2 + 2I$ , justifying your answer. [3]

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- **Example 13 (PYQ M/J 2002/1/1):**

Find the eigenvalues of the matrix

$$\begin{pmatrix} 4 & -2 \\ 3 & -1 \end{pmatrix}.$$

Find also corresponding eigenvectors.

[5]

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- **Example 14 (PYQ O/N 2002/11E):**

The vector  $\mathbf{e}$  is an eigenvector of the square matrix  $\mathbf{G}$ . Show that

- (i)  $\mathbf{e}$  is an eigenvector of  $\mathbf{G} + k\mathbf{I}$ , where  $k$  is a scalar and  $\mathbf{I}$  is an identity matrix,
- (ii)  $\mathbf{e}$  is an eigenvector of  $\mathbf{G}^2$ .

[5]

Find the eigenvalues, and corresponding eigenvectors, of the matrices  $\mathbf{A}$  and  $\mathbf{B}^2$ , where

$$\mathbf{A} = \begin{pmatrix} 3 & -3 & 0 \\ 1 & 0 & 1 \\ -1 & 3 & 2 \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} -5 & -3 & 0 \\ 1 & -8 & 1 \\ -1 & 3 & -6 \end{pmatrix}. \quad [9]$$

**(C) Diagonalisation**

- A  $n \times n$  matrix  $A$  is said to be diagonalisable if there exists a non singular matrix  $P$  and a diagonal matrix  $D$  such that  $P^{-1}AP = D$ .
- This can be rearranged into  $A = PAP^{-1}$  by pre-multiplying both sides with  $P$  and post-multiplying with  $P^{-1}$ .
- This can be used to find power of matrix  $A$ , as if  $A = PDP^{-1}$ , then  $A^n = PD^nP^{-1}$ .

- Also for a diagonal matrix  $D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$ ,  $D^m = \begin{pmatrix} \lambda_1^m & 0 & 0 \\ 0 & \lambda_2^m & 0 \\ 0 & 0 & \lambda_3^m \end{pmatrix}$

**Example 15:**

Matrix  $A$  has eigenvalues 1, 2 with corresponding eigenvectors  $\begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ . Find

- (i)  $A$       (ii)  $A + A^2$       (iii)  $A^n$ , where  $n$  is a positive integer.



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**Case 1 ( $2 \times 2$  matrix):**

- Prove that if  $\lambda_1, \lambda_2$  are the eigenvalues of a  $2 \times 2$  matrix  $A$ , with corresponding eigenvectors  $e_1, e_2$ , then  $AP = PD$ , where  $P$  is the matrix whose columns are  $e_1, e_2$  and  $D$  is the diagonal matrix

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

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**Case 2 ( $3 \times 3$  matrix):**

- Prove that if  $\lambda_1, \lambda_2, \lambda_3$  are the eigenvalues of a  $3 \times 3$  matrix  $A$ , with corresponding eigenvectors  $e_1, e_2, e_3$ , then  $AP = PD$ , where  $P$  is the matrix whose columns are  $e_1, e_2, e_3$  and  $D$  is the diagonal matrix

$$\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}.$$

- **Example 16:**

It is given that  $A = \begin{pmatrix} 2 & 0 \\ -1 & 3 \end{pmatrix}$ . Find a matrix  $P$  and a diagonal matrix  $D$  such that  $A^3 + I = PDP^{-1}$ .

- **Example 17:**

Show that the eigenvalues of  $M = \begin{pmatrix} -3 & a & 2 \\ -2b & 2 & b \\ 0 & 2a & 1 \end{pmatrix}$  are independent of  $a$  and  $b$ .

Obtain, in terms of  $a$  and  $b$ , an eigenvector corresponding to each eigenvalue.

Write down a matrix  $Q$  and a diagonal matrix  $D$  such that  $P = QDQ^{-1}$ .

Hence, for your matrix  $Q$ , evaluate  $M^4Q$  in terms of  $a$  and  $b$ .



- **Example 18:**

The matrix  $A = \begin{pmatrix} 5 & 4 & 1 \\ -6 & -2 & 3 \\ 8 & 8 & 3 \end{pmatrix}$ .

Find a cubic equation whose roots are the eigenvalues of  $A$ , and hence obtain the eigenvalues of  $A$  and a set of corresponding eigenvectors.

Hence find a matrix  $P$  such that  $A = PDP^{-1}$ , where  $D$  is a diagonal matrix which is to be determined.

The matrix  $B$  is such that  $B = QAQ^{-1}$ , where  $Q = \begin{pmatrix} 1 & -2 & 1 \\ 2 & -1 & -1 \\ -1 & 1 & 2 \end{pmatrix}$ .

By using the expression  $PDP^{-1}$  for  $A$ , find the eigenvalues and a set of corresponding eigenvectors of  $B$ .

- **Example 19:**

Find a  $3 \times 3$  matrix  $A$  whose eigenvalues are  $1, -2, -3$ , and for which the eigenvectors are  $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$  respectively.

The element in the first row and the first column of  $A^n$  is denoted by  $a$ .

Show that

$$a = \frac{1}{2} \{(-2)^n + (-3)^n\},$$

and hence find the least integer  $n$  for which  $|a| > 100$ .

- **Example 20 (PYQ O/N 2004/1/12)**

The matrix  $\mathbf{A}$  has  $\lambda$  as an eigenvalue with  $\mathbf{e}$  as a corresponding eigenvector. Show that if  $\mathbf{A}$  is non-singular then

(i)  $\lambda \neq 0$ , [2]

(ii) the matrix  $\mathbf{A}^{-1}$  has  $\lambda^{-1}$  as an eigenvalue with  $\mathbf{e}$  as a corresponding eigenvector. [2]

The matrices  $\mathbf{A}$  and  $\mathbf{B}$  are given by

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & -2 & 4 \\ 0 & 0 & -3 \end{pmatrix} \quad \text{and} \quad \mathbf{B} = (\mathbf{A} + 4\mathbf{I})^{-1}.$$

Find a non-singular matrix  $\mathbf{P}$ , and a diagonal matrix  $\mathbf{D}$ , such that  $\mathbf{B} = \mathbf{PDP}^{-1}$ . [10]

- **Example 21 (PYQ O/N 2005/1/10):**

It is given that the eigenvalues of the matrix  $\mathbf{M}$ , where

$$\mathbf{M} = \begin{pmatrix} 4 & 1 & -1 \\ -4 & -1 & 4 \\ 0 & -1 & 5 \end{pmatrix},$$

are 1, 3, 4. Find a set of corresponding eigenvectors. [4]

Write down a matrix  $\mathbf{P}$  and a diagonal matrix  $\mathbf{D}$  such that

$$\mathbf{M}^n = \mathbf{P}\mathbf{D}\mathbf{P}^{-1},$$

where  $n$  is a positive integer. [2]

Find  $\mathbf{P}^{-1}$  and deduce that

$$\lim_{n \rightarrow \infty} 4^{-n} \mathbf{M}^n = \begin{pmatrix} -\frac{1}{3} & 0 & -\frac{1}{3} \\ \frac{4}{3} & 0 & \frac{4}{3} \\ \frac{4}{3} & 0 & \frac{4}{3} \end{pmatrix}. \quad [5]$$

**(4) Caley-Hamilton Theorem**

- This theorem provides us another way to find powers and inverses of matrices. However, it achieves by establishing a recurrence relation.
- The theorem states: Every matrix satisfies its own characteristics equation.
- For  $3 \times 3$  matrix, if  $\lambda^3 + b\lambda^2 + c = 0$ , then  $M^3 + bM^2 + cI = 0$ .
- **Example 22:**

The matrix  $A$  is given by  $A = \begin{pmatrix} 3 & 1 \\ 5 & -1 \end{pmatrix}$ .

- (a) Find the eigenvalues and corresponding eigenvectors of  $A$ .
- (b) Use the characteristics equation of  $A$  to find the values of  $p$  and  $q$  for which  $A^{-1} = pA + qI$ .

- **Example 23:**

The matrix  $A$  is given by

$$A = \begin{pmatrix} 1 & 3 & 0 \\ 2 & 0 & 2 \\ 1 & 1 & 2 \end{pmatrix}$$

- (a) Find the eigenvalues of  $A$ .
- (b) Use the characteristic equation of  $A$ 
  - (i) to find  $A^{-1}$ .
  - (ii) to obtain an expression for  $A^4$  as a linear combination of  $A$  and  $A^2$ .

- **Example 24:**

The matrix  $M = \begin{pmatrix} 1 & 2 & -2 \\ -4 & 7 & -8 \\ 0 & 0 & -1 \end{pmatrix}$ .

- (i) Find the determinant of this matrix.
- (ii) Find the characteristic equation of the matrix.
- (iii) Find the eigenvalues of this matrix and find the corresponding eigenvectors.
- (iv) Use the characteristic equation to find the inverse of  $M$ .
- (v) Use the characteristic equation to find  $M^4$

- **Example 25:**

The matrix  $A = \begin{pmatrix} -6 & -5 & 1 \\ 24 & 21 & -12 \\ 22 & 20 & -15 \end{pmatrix}$ .

- (a) Show that the eigenvalues of  $A$  are the roots of the equation  $\lambda^3 - 13\lambda + 12 = 0$ . Find these roots and their corresponding eigenvectors.
- (b) Find a matrix  $Q$  and a diagonal matrix  $D$  such that  $A^2 = QDQ^{-1}$ .
- (c) Use the characteristic equation to find  $A^3$ .



- **Example 26:**

The matrix  $M$  is given by

$$M = \begin{pmatrix} 6 & 0 & 0 \\ 4 & -2 & -2 \\ 2 & -4 & 5 \end{pmatrix}$$

- (i) Show that matrices  $Q$  and  $D$  is given by

$$Q = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 2 & 0 & 1 \end{pmatrix} \quad \text{and} \quad D = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

satisfy the equation  $QD = MQ$ .

- (ii) Hence write down the characteristic equation of  $M$ .  
(iii) Prove that  $108M^{-1} = 9M - M^2$  and use this to find  $M^{-1}$ .